Gases

1. A weather balloon is initially filled with 10.0 L helium gas at 1.00 atm and 20.0 °C. The balloon rises to an altitude where the pressure is 26.0 torr and temperature is –50.0 °C. What is the volume (in L) of the balloon at the higher altitude?

2. A 500.0 mL rigid vessel contains 0.240 mol N_2, 0.450 mol H_2, and 0.370 mol O_2 at 298 K. Calculate the partial pressure of each gas and the total pressure (in atm) of the vessel.

3. Assuming *complete* decomposition of 75.0 g $[C(NH_2)_3]NO_3$ {121.11 g/mol} at 1.00 atm and 25.0 °C, what is the total volume (in L) of gases formed by this reaction? Assume temperature and pressure are constant.

$$[C(NH_2)_3]NO_3\ (s) \rightarrow 3\ H_2O\ (g) + 2\ N_2\ (g) + C\ (s)$$

4. Consider the following three containers (**A-C**) filled with gases. Assume all gases are ideal.

A	**B**	**C**
1.0 mol He	1.0 mol N_2	1.0 mol Ar
298 K	298 K	298 K
1.00 L	1.00 L	1.00 L

(a) Which gas has the slowest root-mean-square speed (v_{rms})?

(b) Which gas exhibits the fastest rate of effusion?

(c) Calculate the rates of effusion for He and Ar *relative* to N_2 (i.e., by what factor are these faster or slower?).

Chemical Kinetics

5. Consider the following chemical reaction: **$2\ NOCl\ (g) \rightarrow 2\ NO\ (g) + Cl_2\ (g)$**

   If the rate of disappearance of NOCl (*g*) is 1.0×10^{-3} M/min, determine the rate of appearance of each of the two products in units of M/min.

6. Consider the following aqueous reaction for which initial rate data is collected at 25 °C.

   $Cr(H_2O)_6^{3+}\ (aq) + SCN^-\ (aq) \rightarrow Cr(H_2O)_5SCN^{2+}\ (aq) + H_2O\ (l)$

Expt.	$[Cr(H_2O)_6^{3+}]_0$	$[SCN^-]_0$	Initial Rate
1	1.0×10^{-4} M	0.10 M	2.0×10^{-11} M/s
2	1.0×10^{-3} M	0.10 M	2.0×10^{-10} M/s
3	1.5×10^{-3} M	0.20 M	6.0×10^{-10} M/s
4	1.5×10^{-3} M	0.50 M	1.5×10^{-9} M/s
5	3.0×10^{-3} M	0.25 M	

   (a) Determine the rate law for the reaction, and the value and units of the rate constant (*k*).

   (b) Determine the rate of the reaction (in M/s) for experiment 5 in the table above.

7. Cyclobutene rearranges (isomerizes) into butadiene according to first order kinetics with $k = 2.0 \times 10^{-4}\ s^{-1}$.

cyclobutene → butadiene

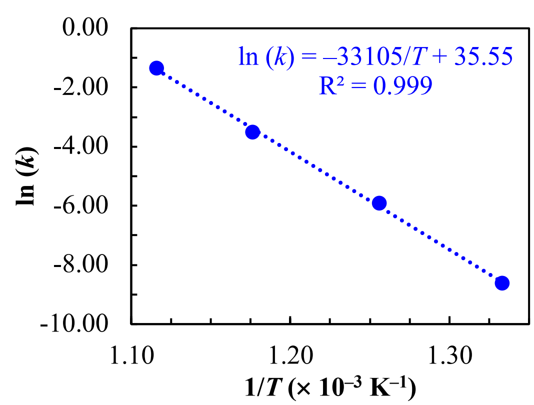
In a flask, 55 torr of cyclobutene is allowed to isomerize. What will be the partial pressure, in torr, of cyclobutene after 30.0 minutes?

8. Plutonium-239 undergoes first-order radioactive decay with a half-life of ^{239}Pu is $t_{1/2} = 24110$ years.

$$^{239}_{94}Pu \rightarrow ^{235}_{92}U + ^{4}_{2}He$$

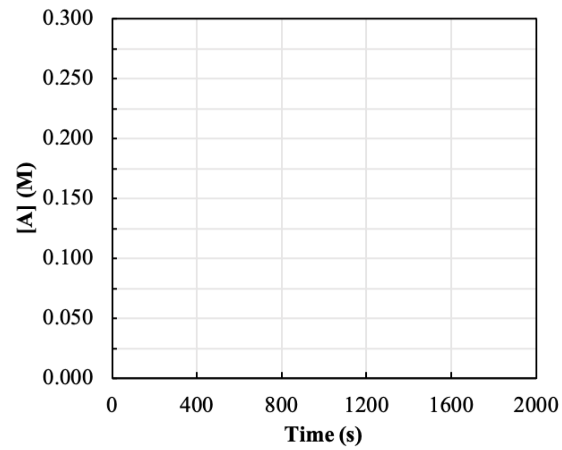
Determine how long (in years) it would take for 65 % of a sample of ^{239}Pu to decay.

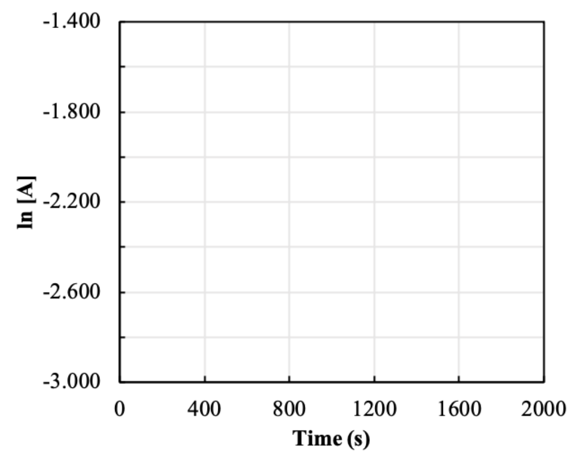
9. Calculate E_a, A, and k at 499 °C using the plot and best-fit line given to the right for an unknown reaction.

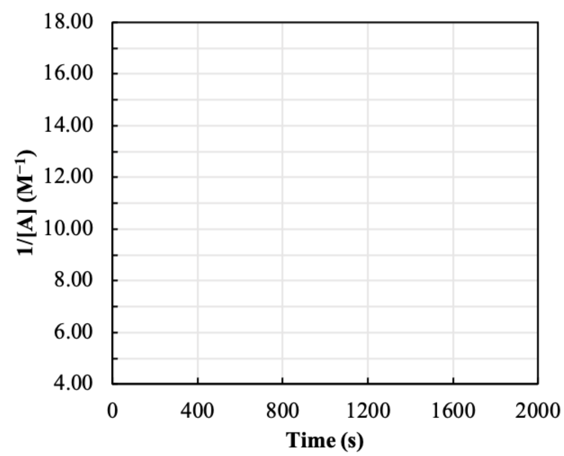


10. Consider the following concentration-time data for some reactant A undergoing some reaction.

Time (s)	**[A] (M)**	**ln[A]**	$\frac{\mathbf{1}}{\mathbf{[A]}}$ $(\mathbf{M^{-1}})$
0	0.240		
400	0.204		
800	0.168		
1200	0.132		
1600	0.096		
2000	0.060		







(a) Fill in the table above by calculating ln[A] and $\frac{1}{[A]}$ for each time given.

(b) Plot [A], ln[A], and $\frac{1}{[A]}$ against time using the plots given to the right. Connect the data points.

(c) The order of the reaction with respect to A is …

0^{th} order 1^{st} order 2^{nd} order

(d) Determine the value and units of the rate constant (k).

11. Given the proposed mechanisms for **Reaction A** below: (i) write the elementary rate law for each elementary step, (ii) write the overall balanced chemical equation, and then (iii) determine the identities of any reactants, products, intermediates, and/or catalysts in the reaction.

	Elementary Step	Elementary Rate Law
step 1	$NO + NO \rightarrow N_2O_2$	
step 2	$N_2O_2 + H_2 \rightarrow H_2O + N_2O$	
step 3	$N_2O + H_2 \rightarrow N_2 + H_2O$	
Overall		

Reactants: ________________ Intermediates: ________________

Products: ________________ Catalysts: ________________

12. If the observed rate law for **Reaction A** is given below, can elementary step 1 be *rate-determining*?

$$\text{Rate} = k[\text{NO}]^2[\text{H}_2]$$

13. The overall decomposition of peroxybenzoic acid in water

$$2\ C_6H_5CO_3H\ (aq) \rightarrow 2\ C_6H_5CO_2H\ (aq) + O_2\ (g)$$

is proposed to occur by the following mechanism.

step 1	$C_6H_5CO_3H(aq) \rightleftharpoons C_6H_5CO_3^-(aq) + H^+(aq)$	(fast)
step 2	$C_6H_5CO_3H(aq) + C_6H_5CO_3^-(aq) \rightarrow C_6H_5CO_2H(aq) + C_6H_5CO_2^-(aq) + O_2(g)$	(slow)
step 3	$C_6H_5CO_2^-(aq) + H^+(aq) \rightarrow C_6H_5CO_2H(aq)$	(fast)

Sketch an energy diagram for the reaction and label the following: reactants, products, intermediates, transition states, and the step with the largest E_a. Assume the product state is higher in energy than the reactant state.

Chemical Equilibrium

14. The chemical equations, equilibrium constants, and initial conditions are given for several reactions. For each, determine the direction in which the system will *shift* to reach equilibrium.

(a) $2\ NH_3\ (g) \rightleftharpoons N_2\ (g) + 3\ H_2\ (g)$ $\quad K_p = 6.8 \times 10^4$

The initial partial pressures are $P_{NH_3,0} = 2.00$ atm, $P_{N_2,0} = 10.00$ atm, and $P_{H_2,0} = 10.00$ atm.

(b) $2\ SO_3\ (g) \rightleftharpoons 2\ SO_2\ (g) + O_2\ (g)$ $\quad K_c = 0.230$

The initial concentrations are $[SO_3]_0 = 2.00$ M, $[SO_2]_0 = 2.00$ M, and $[O_2]_0 = 2.00$ M.

(c) $N_2\ (g) + O_2\ (g) \rightleftharpoons 2NO\ (g)$ $\quad K_c = 0.050$

The initial concentrations are $[N_2]_0 = 0.100$ M, $[O_2]_0 = 0.200$ M, and $[NO]_0 = 1.00$ M.

15. At elevated temperatures, carbon reacts with water vapor according to the equation:

$$C\ (s) + H_2O\ (g) \rightleftharpoons CO\ (g) + H_2\ (g)$$

A reaction mixture contains *only* C and H_2O initially. If $[H_2O]_{eq} = 0.500$ M and $K_c = 0.2$, determine the equilibrium concentration of CO.

16. Consider the oxygen-ozone equilibrium.

$$3\ O_2\ (g) \rightleftharpoons 2\ O_3\ (g) \qquad K_c = 3.8 \times 10^{-13} \text{ at } 2000\text{ K}$$

If you place 1.00 mol O_2 into a 2.00 L vessel at 2000 K and allow the system to reach equilibrium, what will the concentration of O_3 be at equilibrium? **Set up an ICE chart and equilibrium constant expression for this problem, but do not actually solve it.**

17. Iron metal reacts with water vapor to produce iron(III) oxide and hydrogen gas in a vessel.

$$2\ Fe\ (s) + 3\ H_2O\ (g) \rightleftharpoons Fe_2O_3\ (s) + 3\ H_2\ (g)$$

(a) Write the expression for the equilibrium constant, K_c.

(b) Suppose this system is *initially at equilibrium*. If the following changes are made to the equilibrated system, determine the direction in which the reaction will *shift*/proceed to *re*-establish equilibrium.

	Perturbation/Change/Stress			*Shift*		
(i)	More Fe_2O_3 (*s*) is added	*Q*	*K*	Left	No effect	Right
(ii)	Some H_2 (*g*) is removed	*Q*	*K*	Left	No effect	Right
(iii)	More H_2O (*g*) is added	*Q*	*K*	Left	No effect	Right
(iv)	Volume of vessel is doubled	*Q*	*K*	Left	No effect	Right

Acids and Bases

18. Calculate the pH, pOH, [H_3O^+], and [OH^-] of an aqueous solution that is 0.125 M HCl at 25 °C.

19. Consider an aqueous solution that is 0.125 M HCN 25 °C.
    (a) Write the ionization reaction of HCN in water.

    (b) For the reaction above, label both sets of conjugate acid-base pairs.

    (c) Write out the equilibrium constant for the reaction, K_a.

    (d) Do you expect the pH of this solution to be greater than or less than that of 0.125 HCl(*aq*)? Explain.